

9 (**short**) A fibre-reinforced composite consists of aligned carbon fibres in an epoxy resin matrix. The fibres have a Young's modulus n times that of the epoxy matrix E_m . The volume fraction of the fibres is V_f . Using the Materials Data Book expressions for the longitudinal and transverse Young's moduli of a unidirectional fibre-reinforced composite show that:

(a) the longitudinal Young's modulus E_L when the composite is loaded along the fibre direction is given by

$$E_L = E_m(1 + (n-1)V_f) \quad [2]$$

(b) the transverse Young's modulus E_T when the composite is loaded in a direction perpendicular to the fibres is given by

$$E_T = \frac{nE_m}{(n - (n-1)V_f)} \quad [2]$$

(c) the maximum ratio E_L / E_T occurs for $V_f = 0.5$ and is independent of n . [6]